Gromov-Hausdorff-Prokhorov limit and fractal dimension of the leaf-growth measure

Alessandra Caraceni, Nicolas Curien, Robin Stephenson

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Editorial note. Only the selected claim is counted. Other author results are retained as context or local proof support; no mathematical content has been rewritten.

Allowed standard prerequisites (editorial).

- Brownian CRT convergence, Lamperti transforms, subordinator laws of large numbers, the cited bifurcator/spinal decomposition and GHP/Prokhorov foundational results. Standard cited foundations only; all selected local constructions and proof estimates remain in source ranges.

In the case $d = 2$ of plane binary trees, however, such a coupling between τ_n and τ_{n+1} – where d is henceforth omitted – can be made very explicit (and in fact it is unique provided some natural symmetries are enforced). Given a uniform plane binary tree τ_n of *size* n (i.e., with n vertices that are *not* leaves, and hence with $n+1$ leaves), one can produce an explicit measure ν_{τ_n} on the set of the $n+1$ leaves of τ_n with the following property. If, conditionally on τ_n, one samples l according to ν_{τ_n}, then the tree $\mathrm{grow}(\tau_n, l)$ obtained by attaching a "cherry" (two sister leaves) to l is uniformly distributed among all plane binary trees with $n+1$ vertices. See Figure 2.

Construction of the leaf-growth measures

The (random) measure ν_{τ_n}, which we call the *leaf-growth* measure, is explicit as a function of τ_n. Precisely, given $\tau_n = t$, and given any leaf l of t, the mass given to l is

$$\nu_t(l) = \prod_{i=1}^{|l|} C(a_i, b_i), \tag{1}$$

where

- $|l|$ is the height of l in t;
- if v_i is the ancestor of l at height i, then a_i is the size of the subtree of descendants of v_i (including v_i itself);
- $b_i = a_{i-1} - a_i - 1$ (setting $a_0 = n$) is the size of the subtree of descendants of the *sibling* of v_i;
- $C(a, b)$ is the explicit rational function

$$C(a,b) := \frac{(a+1)(2a+1)(a+3b+3)}{(a+b+1)(a+b+2)(2(a+b)+3)}.$$

0.1 Some useful asymptotics

Before we move on to investigate several aspects of the leaf-growth measures ν_t, it is convenient to state some of the main (easy or well known) approximation results we will need in what follows, both for the quantities $C(a, b)$ and the probabilities $P(a, b)$.

Note that, fixing $b = n - 1 - a$, we can write out the values of $P(a, b)$ as a quotient of factorials

$$P(a,b) = \frac{(2a)!(2b)!n!(n+1)!}{a!(a+1)!b!(b+1)!(2n)!}.$$

Using Stirling's approximation, it is clear that, for $x \in (0,1)$,

$$4\sqrt{\pi} \lim_{\substack{n\to\infty \\ a/n\to x}} n^{3/2} P(a,b) = \frac{1}{x^{3/2}(1-x)^{3/2}} =: \mathsf{p}(x); \tag{2}$$

moreover, for all $x \in [0,1]$,

$$\lim_{\substack{n\to\infty \\ a/n\to x}} C(a,b) = x^2(3-2x) =: \mathsf{c}(x). \tag{3}$$

The more compact notation $\mathsf{p}(x)$ and $\mathsf{c}(x)$ for the functions $x^{-3/2}(1-x)^{-3/2}$ and $x^2(3-2x)$ (see Figure 1) that is being introduced here will be used frequently in future sections.

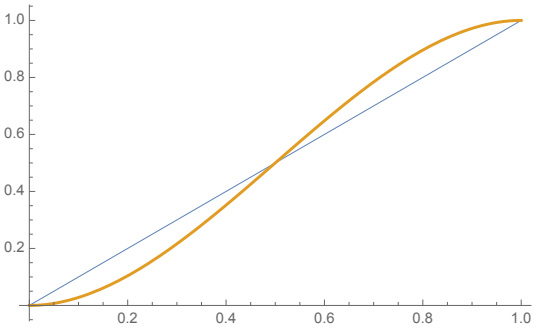


Figure 1: A plot of the function $x \mapsto \mathsf{c}(x)$. Notice that it is symmetric with respect to $(1/2, 1/2)$ and below the first bisector when $x \in [0, 1/2]$.

1 The leaf-growth measure on the Brownian CRT

In this section we construct the leaf-growth measure $\nu_{\mathcal{T}}$ on the Brownian CRT and prove that (a scalar multiple of) the latter, equipped with the former, is the scaling limit of the discrete model for the Gromov–Hausdorff–Prokhorov metric. In order to establish the scaling limit, we provide an explicit description of the spine towards a typical $\nu_{\mathcal{T}}$-point.

1.1 Preliminary notation

The Brownian Continuum Random Tree (CRT) is the random real tree $(\mathcal{T}, d)$ coded by a standard Brownian excursion (length 1, variance 1). This differs by a factor $1/2$ compared to Aldous' original definition. We refer to [32] for its main properties and for standard notation about real trees, some of which is recalled below. It is given with a distinguished point ρ called the *root* of the tree. The degree of a point $x \in \mathcal{T}$ is the number of connected components of $\mathcal{T} \setminus \{x\}$. When the degree is equal to 1 we say x is a *leaf*; we call x a *point of the skeleton* when its degree is 2 and a *branch point* when its degree is 3 (there are almost surely no points with degree 4 or more in the Brownian CRT). For $x \in \mathcal{T}$, we call *height* of x its distance to ρ and denote it by $\mathrm{ht}(x)$. We say that x is a *descendant* of another point $y \in \mathcal{T}$ (and equivalently that y is an *ancestor* of x) if y lies on the segment between ρ and x, and we call $\mathcal{T}_x$ the subset of $\mathcal{T}$ formed by x and all its descendants. Finally, for $x \in \mathcal{T}$ and $t \geq 0$ we define for the point $[x]_t$ as:

- the unique ancestor of x with height t, if $t < \mathrm{ht}(x)$
- x itself, if $t \geq \mathrm{ht}(x)$.

In particular, $\mathcal{T}_{[x]_t}$ denotes the subtree above height t which contains the point x. When $[x]_t$ is a branch point and if $t < \mathrm{ht}(x)$, then $\mathcal{T}_{[x]_t} \setminus \{[x]_t\}$ has two components and we denote by $\mathcal{T}_{[x]_{t+}}$ the closure of component that contains x.

The Brownian CRT is also endowed with its uniform mass measure μ (also called the leaf measure), which can either be seen as the projection of the Lebesgue measure on $[0,1]$ in the coding by the Brownian excursion, or more intrinsically as its natural Hausdorff measure, see [17, 18].

In our work, another description of $\mathcal{T}$ will be of key importance. Namely, the Brownian CRT can (with this normalisation) be seen as a fragmentation tree in the sense of Haas and Miermont [25] with self-similarity index $\alpha = -1/2$ for the dislocation measure

$$\mathsf{Dis} := \sqrt{\frac{2}{\pi}} \int_{1/2}^{1} \frac{\mathrm{d}x}{(x(1-x))^{3/2}}\, \delta_{(x,(1-x))}. \tag{4}$$

Heuristically, this means that it encodes the genealogy of a system of particles starting from a single particle of mass 1, where particles of mass m evolve independently and split into two particles of mass mx and $m(1-x)$ at rate $m^{-1/2} \cdot \sqrt{2/\pi}\, \mathsf{p}(x)$, where $\mathsf{p}(x)$ is as in (2). Following Bertoin [8], those particles alive at time t correspond to the subtrees of $\mathcal{T}$ above height t and their masses correspond to the μ-masses of those subtrees.

1.2 Definition of ν

Conditionally on the CRT $\mathcal{T}$, the leaf-growth measure $\nu_{\mathcal{T}}$, which throughout this section we will also denote by ν, rendering the dependence on $\mathcal{T}$ implicit, will be defined by specifying the value $\nu(\mathcal{T}_x)$ for all non-leaf $x \in \mathcal{T}$. Proposition 1 of [42] will then guarantee its existence and uniqueness. The definition will be analogous to the discrete case: there, the relative masses given to the subtrees in an (a, b) split were $C(a, b)$ and $C(b, a)$, and here the relative masses given to the subtrees in a $(x, 1-x)$ split will be $\mathsf{c}(x)$ and $1 - \mathsf{c}(x) = \mathsf{c}(1-x)$.

Note that the construction is purely deterministic: for the purposes of Proposition 2.4, one can take $\mathcal{T}$ to be any binary compact rooted real tree equipped with a measure μ supported by the leaves, and c to be any symmetric function satisfying $\mathsf{c}(x) \leq x$ for $x \in [0, 1/2]$. Since our only application concerns the Brownian CRT with the specific function $\mathsf{c} = x^2(3-2x)$, for the sake of simplicity we shall keep using the the notations $\mathcal{T}$, μ and c throughout this section, though they should be interpreted here with this more general framework in mind.

Given $x \in \mathcal{T}$, recall from above that we denote by $\mathcal{T}_{[x]_t}$ and $\mathcal{T}_{[x]_{t+}}$, respectively, the subtree above height t which contains x, and in the case of a branch point, the closure of the connected component of $\mathcal{T} \setminus [x]_t$ which contains x. Finally we set

$$u_x(t) := \frac{\mu(\mathcal{T}_{[x]_{t+}})}{\mu(\mathcal{T}_{[x]_t})} \in [0, 1].$$

In particular, since the tree $\mathcal{T}$ is binary and compact, there are for any $x \in \mathcal{T}$ at most countably many branch points on the ancestral path of x, hence we have $u_x(t) < 1$ only for a countable number of heights $t \leq \mathrm{ht}(x)$. In particular, if $(t_i : i \in \mathbb{N})$ is an enumeration of the heights of those branch points then the family $(\mu(\mathcal{T}_{[x]_{t_i}}) - \mu(\mathcal{T}_{[x]_{t_i+}}) : i \in \mathbb{N})$ is summable, and since the measure μ does not charge the skeleton, its sum is equal to $1 - \mu(\mathcal{T}_x)$. We deduce then that for all $x \in \mathcal{T}$ we have

$$\mu\big(\mathcal{T}_x\big) = \prod_{0 \leq t < \mathrm{ht}(x)} u_x(t). \tag{5}$$

1.3 Path towards a typical ν-leaf and spine decomposition

Recalling the definition of the Brownian dislocation measure in (4), we now build two measures π_μ and π_ν on $(0, \infty)$ defined by

$$\begin{aligned}\int_0^\infty \mathrm{d}\pi_\mu(s)\ \phi(s) &:= \sqrt{\frac{2}{\pi}} \int_{1/2}^{1} \big(\mathsf{c}(x)\phi(-\log(x)) + \mathsf{c}(1-x)\phi(-\log(1-x))\big)\mathsf{p}(x)\mathrm{d}x \\ &= \sqrt{\frac{2}{\pi}} \int_0^1 \mathsf{c}(x)\phi\left(-\log(x)\right)\mathsf{p}(x)\mathrm{d}x\end{aligned}$$

and

$$\begin{aligned}\int_0^\infty \mathrm{d}\pi_\nu(s)\ \phi(s) &:= \sqrt{\frac{2}{\pi}} \int_{1/2}^{1} \big(\mathsf{c}(x)\phi(-\log(\mathsf{c}(x))) + \mathsf{c}(1-x)\phi(-\log(\mathsf{c}(1-x)))\big)\mathsf{p}(x)\mathrm{d}x \\ &= \sqrt{\frac{2}{\pi}} \int_0^1 \mathsf{c}(x)\phi\left(-\log(\mathsf{c}(x))\right)\mathsf{p}(x)\mathrm{d}x.\end{aligned}$$

Consider then Π_μ a Poisson point process (P.p.p.) on $\mathbb{R}_+\times\mathbb{R}_+$ with intensity $\mathrm{d}\pi_\mu\otimes\mathrm{d}t$. By replacing each atom (p,t) of Π_μ by the atom $(-\log(\mathsf{c}(\mathrm{e}^{-p})),t)$ we obtain another Poisson point process with intensity $\mathrm{d}\pi_\nu\otimes\mathrm{d}t$. We can then define two coupled (non-decreasing) subordinators ξ_μ and ξ_ν started from 0, and with respective Lévy measures π_μ and π_ν, and obtained by summing the atoms of the respective P.p.p., i.e.,

$$\xi_\mu(t)=\sum_{\substack{(p,s)\in\Pi_\mu\\ s\le t}} p \quad\text{and}\quad \xi_\nu(t)=\sum_{\substack{(p,s)\in\Pi_\nu\\ s\le t}} p=\sum_{\substack{(p,s)\in\Pi_\mu\\ s\le t}} -\log(\mathsf{c}(\mathrm{e}^{-p})). \tag{6}$$

Then we perform the Lamperti transformation with index $1/2$ using ξ_μ : we let

$$\theta(t)=\inf\Big\{u\ge 0:\int_0^u \mathrm{e}^{-\frac{1}{2}\xi_\mu(r)}\mathrm{d}r>t\Big\}, \tag{7}$$

and build two processes X_μ (a so-called positive self-similar Markov process, see [30]) and X_ν (which is not Markovian, since the time change uses information from X_μ) defined for $t\ge 0$ by

$$(X_\mu(t),X_\nu(t)):=(\mathrm{e}^{-\xi_\mu(\theta(t))},\mathrm{e}^{-\xi_\nu(\theta(t))}). \tag{8}$$

Both of those processes are pure jump and non-increasing from 1 to 0, and are absorbed at 0 at time

$$I:=\int_0^\infty \mathrm{e}^{-\frac{1}{2}\xi_\mu(t)}\mathrm{d}t \tag{9}$$

which is almost surely finite since ξ_μ drifts to $+\infty$ at linear speed.

1.4 Scaling limits

The aim of this section is to prove Theorem 2.3. For the reader's convenience, we quickly recall the definition of the (pointed) Gromov–Hausdorff–Prokhorov convergence and refer to [2, 28] for more details, as well as [6] for a presentation of the topology without the metric. A weighted pointed compact metric space (M,d,ρ,μ) is a compact metric space equipped with a finite Borel measure μ and a distinguished point $\rho\in M$. We let $\mathbb{M}$ be the set of all isometry classes of weighted pointed compact metric spaces. To lighten notation, we shall often identify a compact weighted metric space with its equivalence class. We equip $\mathbb{M}$ with the classical Gromov–Hausdorff–Prokhorov metric, defined for every $\boldsymbol{M}:=(M,d,\rho,\mu)$ and $\boldsymbol{M'}:=(M',d',\rho',\mu')$ in $\mathbb{M}$ by:

$$d_{\mathrm{GHP}}\big(\boldsymbol{M},\boldsymbol{M'}\big):=\inf_{\phi,\phi'}\Big(\delta_{\mathrm{H}}\big(\phi(M),\phi'(M')\vee\delta_{\mathrm{P}}\big(\phi_*\mu,\phi'_*\mu'\big)\vee\delta(\phi(\rho),\phi'(\rho')\Big),$$

where the infimum is taken over all isometries ϕ, ϕ' from M, M' into a metric space (Z,δ) and δ_{H} (resp. δ_{P}) stands for the classical Hausdorff distance (resp. the Prokhorov distance) in Z. The space $(\mathbb{M},d_{\mathrm{GHP}})$ is then a Polish space.

Our starting point is the well-known convergence in distribution from [5]:

$$\Big(\frac{1}{\sqrt{n}}\tau_n,\rho_n,\mu_n\Big)\xrightarrow[n\to\infty]{(d)}(2\sqrt{2}\,\mathcal{T},\rho,\mu), \tag{10}$$

for the pointed Gromov–Hausdorff–Prokhorov metric, where μ_n is the uniform measure on the vertices of τ_n. By analogy with this notation, we shall also use the notation ν_n for the measure ν_{τ_n} introduced in Section 2. We saw in Section 2 and in Proposition 2.4 that the measures ν_n and ν are obtained as measurable functions of $((1/\sqrt{n})\tau_n,\rho_n,\mu_n)$ and $(2\sqrt{2}\,\mathcal{T},\rho,\mu)$ respectively. Furthermore, both measures are constructed similarly by modifying the splitting of the reference measures μ_n and μ using the functions $C(a,b)$ and c. In view of (10) and the convergence (3), it is natural to expect that those functions are almost "continuous", letting us adjoin ν_n and ν to (10) using deterministic arguments.

2 Building schemes for trees: leaf-growth measures

In this section, we set up some notation and give some context for what we are about to discuss, i.e., how to "uniformly grow" trees "from their leaves". For completeness, we shall state some known results in their general form, which involves d-ary trees (see Luczak & Winkler [36]).

In the case $d=2$, motivated by applications to the study of mixing times of flip chains, this procedure has been made completely explicit by Caraceni and Stauffer [15] and forms the basis of the current work.

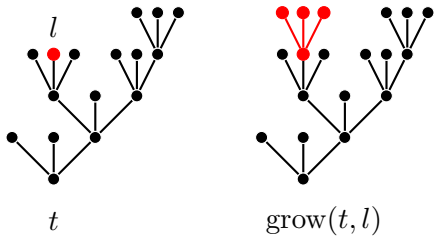


Figure 2: A plane ternary tree of size 5 (i.e., with 5 internal vertices and thus $3 + 2 \cdot 4 = 11$ leaves) is grown at the leaf l by adding 3 children to l, thus creating the tree $\mathrm{grow}(t, l)$ of size 6.

2.1 Building schemes for d-ary trees

Definition 2.1. Let $d \geq 2$ and $n \geq 0$ be integers. We shall call a *d-ary tree of size n* a rooted plane tree whose vertices all have d children (we call such vertices *internal*) or no children (we call such vertices *leaves*).

We say that a d-ary tree τ has *size* $|\tau| = n$ if it has n internal vertices. For simplicity, we shall commit a slight abuse of notation and also write τ for the set of vertices of τ, so we can write $v \in \tau$ to mean that v is a vertex of τ.

We call $\mathsf{T}_n^{(d)}$ the set of all d-ary trees of size n and write $\mathsf{T}^{(d)}$ for $\bigcup_{n \geq 0} \mathsf{T}_n^{(d)}$. Note that the set $\mathsf{T}_0^{(d)}$ contains one element, the tree consisting of only its root.

A d-ary tree has a natural recursive structure: if $t \in \mathsf{T}_n^{(d)}$ with $n \geq 1$, it naturally induces a sequence of d-ary trees $(t^1, \ldots, t^d)$: letting $\rho \in t$ be the root, one can erase all children of ρ other than the ith child v_i and consider the connected component of v_i, rooted at v_i, to obtain t^i. One then has $\sum_{i=1}^{d} |t^i| = n - 1$. From this decomposition one obtains the *profile* of the tree t:

Definition 2.2. Given $t \in \mathsf{T}_n^{(d)}$ with $n \geq 1$, its *profile* is the vector $\underline{x}(t) = (|t^1|, \ldots, |t^d|) \in \mathbb{N}^d$.

Note that the term profile is often used with different meanings (for example, for the sequence of generation sizes) in similar contexts; however, we choose to employ it here to keep notation at least partially consistent with the original paper [36].

We now discuss a particular "growth" operation on d-ary trees. Given a pair (t, l), where t is a d-ary tree of size n and l is a leaf of t, we construct a tree $\mathrm{grow}(t, l)$ of size $n + 1$ by adding d leaves to t as children of the vertex l, thus turning l into an internal vertex.

Luczak and Winkler consider the following natural question in [36]: supposing τ_n is a uniform random element of $\mathsf{T}_n^{(d)}$ and τ_{n+1} is a uniform random element of $\mathsf{T}_{n+1}^{(d)}$, is it possible to couple τ_n and τ_{n+1} in such a way that τ_{n+1} is always of the form $\mathrm{grow}(\tau_n, l_n)$, where l_n is some leaf of the tree τ_n?

They call a family of such couplings for all $n \geq 0$ a *building scheme* for d-ary trees and answer the question in the affirmative, thus proving

Theorem 2.3. *We have the following convergence in distribution for the Gromov–Hausdorff–Prokhorov topology (see Section 1.4 for a definition):*

$$\left(n^{-1/2} \cdot \tau_n, \nu_{\tau_n}\right) \xrightarrow[n \to \infty]{(d)} (2\sqrt{2}\,\mathcal{T}, \nu_{\mathcal{T}}),$$

where $\mathcal{T}$ is the Brownian CRT and $\nu_{\mathcal{T}}$ is a probability measure on the leaves of $\mathcal{T}$ with Hausdorff dimension $2\gamma = 6(2 - \sqrt{3}) \approx 1.6077$.

Proposition 2.4 (Definition of ν)**.** *Suppose that $\mathcal{T}$ is a compact binary rooted tree equipped with a measure μ supported by its leaves. If $\mathsf{c} : [0,1] \to [0,1]$ is symmetric with respect to $1/2$ and satisfies $\mathsf{c}(x) \leq x$ for $x \in [0, 1/2]$, then there exists a unique Borel measure ν supported by the leaves of $\mathcal{T}$ specified by*

$$\nu(\mathcal{T}_x) := \prod_{0 \leq t < \mathrm{ht}(x)} \mathsf{c}(u_x(t)). \tag{11}$$

Proof. Let, for $x \in \mathcal{T}$, $m(x) = \prod_{0 \le t < \mathrm{ht}(x)} \mathsf{c}(u_x(t))$. Since we have $\mathsf{c}(x) \le x$ for $x \in [0, 1/2]$ the product is well-defined and is non-zero as soon as $\prod_{0 \le t < \mathrm{ht}(x)} u_x(t)$ is non-zero. One can straightforwardly check the following:

- m is a well-defined and nonnegative function, which is equal to 0 exactly on the leaves of $\mathcal{T}$.
- m is non-increasing, in the sense that, if x is an ancestor of y, then $m(x) \ge m(y)$.
- m is "left-continuous", in the sense that $m(x) = \lim_{s \to (\mathrm{ht}(x))^-} m([x]_s)$.
- m is "additively right-continuous": if x is not a branch point, then m is continuous at x, while if x is a branch point, then, letting y and z be two descendants of x not in the same one of the two subtrees originating at x, then $m(x) = \lim_{s \to (\mathrm{ht}(x))^+} m([y]_s) + m([z]_s)$.

These facts, combined, let us use [42, Prop. 1], which then tells us there exists a unique measure ν such that $\nu(\mathcal{T}_a) = m(a)$ for a, and that it has no atoms. With the notation introduced just before the Proposition, since μ does not charge the skeleton, we can also check that $1 - \nu(\mathcal{T}_x) = \sum_{i \in \mathbb{N}} \big(\nu(\mathcal{T}_{[x]_{t_i}}) - \nu(\mathcal{T}_{[x]_{t_i+}})\big)$, in particular ν gives no mass to the ancestral path of x, and as a result is supported by the leaves of $\mathcal{T}$. □

Theorem 2.5 (Spine decomposition)**.** *Conditionally on the Brownian CRT $\mathcal{T}$, let L be a ν-distributed random point of $\mathcal{T}$. Then the bivariate process*

$$(\mu(\mathcal{T}_{[L]_{t+}}), \nu(\mathcal{T}_{[L]_{t+}}))_{t \ge 0}$$

(under the annealed law) is distributed as (X_μ, X_ν). As a consequence, L is almost surely a leaf, and its height has the law of I.

Moreover, let $(t_i, i \in \mathbb{N})$ be the times at which $u_L(\cdot) \in (0, 1)$, and for all $i \in \mathbb{N}$ let $\mathcal{T}_i$ be the subtrees which branch off $[\![\rho, L]\!]$ at those times (ranked, say in decreasing order of their μ-masses). Then, conditionally on the whole process $(\mu(\mathcal{T}_{[L]_t}), t \ge 0)$, the trees

$$\big(u_L(t_i) \cdot \mu(\mathcal{T}_{L(t_i)})\big)^{-1/2} \cdot \mathcal{T}_i, \quad i \ge 1,$$

are independent Brownian CRT's.

Proof. Informally, the idea is that the path to L encounters binary splits at the same rate as the path to a uniform leaf, but crosses those splits differently, using the probability c. This is formalised in [40] by the notion of *bifurcator*. If L' is a leaf with distribution μ, then the paths from ρ to L and to L' will encounter countable numbers of binary branch points. At any common such branch point, let x and $1 - x$ be the relative μ-masses of both pending subtrees, choosing x to be the mass of the one containing L. Then the probability that L' is in the other subtree, known as the "switching probability" in [40], is equal to $\mathsf{c}(1 - x)$. While the process $(Y(t), t \ge 0)$ defined by $Y(t) = \mu(\mathcal{T}_{[L']_{t+}})$ is well-known [8, §4] to be the Lamperti transform of a subordinator with Lévy measure π_{unif}, defined by

$$\begin{aligned}\int_0^\infty \mathrm{d}\pi_{\mathrm{unif}}(s)\ \phi(s) &= \sqrt{\frac{2}{\pi}} \int_{1/2}^1 \big(x\phi(-\log(x)) + (1-x)\phi(-\log(1-x))\big)\mathsf{p}(x)\mathrm{d}x \\ &= \sqrt{\frac{2}{\pi}} \int_0^1 x\phi(-\log(x))\mathsf{p}(x)\mathrm{d}x,\end{aligned}$$

Proposition 1 from [40] states then that the process $\mu(\mathcal{T}_{[L]_{t+}}), t \ge 0$ will be the Lamperti transform of a subordinator where the initial x in the integrand of the above is replaced by $\mathsf{c}(x)$, giving us π_μ.

That $(\nu(\mathcal{T}_{[L]_{t+}}), t \ge 0)$ has the appropriate paired distribution comes from the fact that its multiplicative jumps are obtained by applying c to those of $(\mu(\mathcal{T}_{[L]_{t+}}), t \ge 0)$, which is how X_ν is derived from X_μ.

Finally, the spinal decomposition itself is again a consequence of [40], this time Lemma 21. □

Proposition 2.6. *Almost surely for $\mathcal{T}$ and a.e. for $\nu(\mathrm{d}\ell)$ we have*

$$\lim_{r \to 0} \frac{\log \nu\big(B_r(\ell)\big)}{\log r} = 2\gamma.$$

Proof. Recall that $\gamma = 3(2-\sqrt{3})$, and the definitions of $\mathsf{p}(x)$ in (2), $\mathsf{c}(x)$ in (3), ξ_μ and ξ_ν in (6), X_μ and X_ν in (8), the time change θ in (7) and extinction time I in (9). By Theorem 2.5, we can assume that the bivariate process $(\mu(\mathcal{T}_{[L]_{t+}}), \nu(\mathcal{T}_{[L]_{t+}}))_{t\geq 0}$ is given by (X_μ, X_ν).

The local rate of growth of ν near a ν-typical point is related to the asymptotic behaviour of X_μ and X_ν near their extinction time. To understand this, let us compute the first moments of the Lévy processes ξ_μ and ξ_ν : we have

$$\mathbb{E}[\xi_\mu(t)] = t\int_0^\infty s\ \mathrm{d}\pi_\mu(s) = \sqrt{2\pi} \quad \text{and} \quad \mathbb{E}[\xi_\nu(t)] = t\int_0^\infty s\ \mathrm{d}\pi_\nu(s) = \gamma\sqrt{2\pi}.$$

Since the Lévy measure integrates the identity function, by the law of large numbers we have $t^{-1}\xi_\mu(t) \to \sqrt{2\pi}$ and $t^{-1}\xi_\nu(t) \to \gamma\sqrt{2\pi}$ as $t \to \infty$ almost surely. Through the Lamperti transformation (8), this turns into estimate on the time change θ since for $\varepsilon > 0$,

$$\int_{\theta(I-\varepsilon)}^\infty \mathrm{e}^{-\frac{1}{2}\xi_\mu(t)}\mathrm{d}t = \varepsilon,$$

from which we straightforwardly deduce using the above law of large numbers that

$$\theta(I-\varepsilon) \underset{\varepsilon\to 0}{\sim} -\frac{2}{\sqrt{2\pi}}\log\varepsilon$$

almost surely. Substituting back into (8), we find using the same law of large numbers that

$$\lim_{\varepsilon\to 0}\frac{\log X_\mu(I-\varepsilon)}{\log\varepsilon} = 2 \quad \text{and} \quad \lim_{\varepsilon\to 0}\frac{\log X_\nu(I-\varepsilon)}{\log\varepsilon} = 2\gamma,$$

also almost surely.

Now, by the first item of the spine decomposition (Theorem 2.5), we clearly have $\nu\big(B_\varepsilon(L)\big) \leq X_\nu(I-\varepsilon)$ as a process in ε and together with the last display this entails that

$$\liminf_{\varepsilon\to 0}\frac{\log\nu\big(B_\varepsilon(L)\big)}{\log\varepsilon} \geq 2\gamma, \quad \text{a.s..}$$

Let us now prove the upper bound for the dimension via a lower bound on $\nu\big(B_\varepsilon(L)\big)$. For this, we concentrate on jumps of ξ_μ of size at least $\log 2$. Remember that thanks to the spine decomposition, conditionally on such a jump of size $s \geq \log 2$ appearing at time t, the subtree $\mathcal{T}_{[L]_u} \setminus \mathcal{T}_{[L]_{u+}} \cup \{[L]_u\}$ with $t = \theta(u)$ (which is the subtree which is "left on the side" at height u) is an independent Brownian CRT with μ-mass

$$(1-\mathrm{e}^{-s})\mathrm{e}^{-\xi_\mu(t-)} \geq \frac{1}{2}\,\mathrm{e}^{-\xi_\mu(t-)}.$$

Notice two things: first, the ν-mass of this subtree is at least $\mathrm{e}^{-\xi_\nu(t-)}/2$ (because $\mathsf{c}(x) \leq x$ for $x \in [0,1/2]$) and second, this tree has a fixed probability equal to $\mathbb{P}(\mathrm{ht}(\mathcal{T}) \leq 1)$ of being of height less than the square root of its μ-mass. Under those conditions, this subtree is a subset of $B_\varepsilon(L)$ for $\varepsilon \geq \mathrm{e}^{-\frac{1}{2}\xi_\mu(t-)} + \int_t^\infty \mathrm{e}^{-\frac{1}{2}\xi_\mu(s)}\mathrm{d}s$ and we have

$$\nu\big(B_\varepsilon(L)\big) \geq \frac{1}{2}\,\mathrm{e}^{-\xi_\nu(t-)}. \tag{12}$$

By the law of large numbers on ξ_μ and ξ_ν, we can take $\varepsilon = c_1\mathrm{e}^{-t\sqrt{2\pi}/2}$ and we have $\nu\big(B_\varepsilon(L)\big) \geq c_2\mathrm{e}^{-t\sqrt{2\pi}\gamma/2} = \varepsilon^{2\gamma+o(1)}$ for some $c_1, c_2 > 0$. The preceding inequality is only valid for those ε associated to a jump time t satisfying the above conditions. However, if we can find a family $t_i \to \infty$ of such times satisfying

$$0 < t_1 < t_2 < \cdots < t_i \xrightarrow[i\to\infty]{} \infty \quad \text{and} \quad \lim_{t\to\infty}\frac{t_{i+1}}{t_i} = 1, \tag{13}$$

then the bound $\nu\big(B_\varepsilon(L)\big) \geq \varepsilon^{2\gamma+o(1)}$ still holds as $\varepsilon \to 0$ and proves the lower bound on the Hausdorff dimension. To see (13), notice that in the time parametrisation of ξ_ν, the intensity $C > 0$ of such times is positive: by a standard law of large numbers we can almost surely find a diverging sequence of jump times t_i satisfying all of the above assumptions and which satisfy $t_i \sim Ci$. This entails (13) and completes the proof. □

Proof of Theorem 2.3. Using the Skorokhod representation theorem, we can assume that (10) holds almost surely, and the proof will proceed deterministically. Next, by a general representation theorem for Gromov-Hausdorff-type topologies [29, Lem. 2.5], we can embed all trees τ_n and $\mathcal{T}$ isometrically in the same compact metric space (E, d_E)—we keep the same notation for simplicity—, in such a way that

$$\Big(\frac{1}{\sqrt{n}}\tau_n, \rho_n, \mu_n\Big) \xrightarrow[n\to\infty]{} (2\sqrt{2}\,\mathcal{T}, \rho, \mu),$$

holds in pointed Hausdorff-Prokhorov sense inside E. Fix then any point x in the skeleton of $\mathcal{T}$ which is not a branch point. Lemma 3.1 shows that there exists a sequence $(x_n, n \in \mathbb{N})$ with $x_n \in \tau_n$ such that $x_n \to x$ in E and $\mu_n((\tau_n)_{x_n}) \to \mu(\mathcal{T}_x)$. We will now show that we also have

$$\lim_{n\to\infty} \nu_n((\tau_n)_{x_n}) = \nu(\mathcal{T}_x), \tag{14}$$

which will guarantee via the Portmanteau theorem and [42, Prop. 1], that any subsequential limit of (ν_n) is equal to ν, ending the proof of Theorem 2.3 since $\mathcal{T}$ is compact.

Let us denote by h_n the height of x_n and by $(a_i^{(n)}, b_i^{(n)})$ the sizes of the two subtrees above the ancestor of x_n at height $0 \le i < h_n$, with the convention that $a_i^{(n)}$ is the size of the subtree containing x_n. It is then straightforward, but rather long, to prove that we also have the convergence

$$\left(\Big(\frac{a_i^{(n)}}{a_i^{(n)}+b_i^{(n)}}, \frac{i}{2\sqrt{2n}}\Big) : 0 \le i < h_n\right) \xrightarrow[n\to\infty]{} \left(\Big(\frac{\mu(\mathcal{T}_{[x]_{t+}})}{\mu(\mathcal{T}_{[x]_t})}, t\Big) : 0 \le t < \mathrm{ht}(x)\right) \tag{15}$$

in terms of point processes on $\mathbb{R}_+^* \times \mathbb{R}_+$. In particular, recall that (5) holds almost surely. Recall first from the construction of ν_n and μ_n that we have

$$\nu_n((\tau_n)_{x_n}) = \prod_{i=0}^{h_n-1} C(a_i^{(n)}, b_i^{(n)}) \quad \text{and} \quad \mu_n((\tau_n)_{x_n}) = \prod_{i=0}^{h_n-1} \frac{a_i^{(n)}}{a_i^{(n)}+b_i^{(n)}+1}. \tag{16}$$

On the continuous side, if we set $a_t = \mu(\mathcal{T}_{[x]_{t+}})$ and $b_t = \mu(\mathcal{T}_{[x]_t}) - \mu(\mathcal{T}_{[x]_{t+}})$ for $0 \le t \le \mathrm{ht}(x)$ (note that $b_t = 0$ when $[x]_t$ is not a branch point), then by (5) and Proposition 2.4 we have

$$\nu(x) = \prod_{t<\mathrm{ht}(x)} \mathsf{c}(a_t/(a_t+b_t)) \quad \text{and} \quad \mu(x) = \prod_{t<\mathrm{ht}(x)} \frac{a_t}{a_t+b_t}. \tag{17}$$

On the one hand, the convergences (15), (3) and Fatou's lemma directly imply that

$$\limsup_{n\to\infty} \nu_n((\tau_n)_{x_n}) = \limsup_{n\to\infty} \prod_{i=0}^{h_n-1} C(a_i^{(n)}, b_i^{(n)}) \le \prod_{t<\mathrm{ht}(x)} \mathsf{c}(a_t/(a_t+b_t)) = \nu(x).$$

To get the matching lower bound $\nu(x) \le \liminf_{n\to\infty} \nu_n((\tau_n)_{x_n})$, notice that the same argument yields

$$\limsup_{n\to\infty} \mu_n((\tau_n)_{x_n}) = \prod_{i=0}^{h_n-1} \frac{a_i^{(n)}}{a_i^{(n)}+b_i^{(n)}+1} \le \mu(x) = \prod_{t<\mathrm{ht}(x)} \frac{a_t}{a_t+b_t};$$

here, however, we know that the matching lower bound holds since

$$\lim_{n\to\infty} \mu_n((\tau_n)_{x_n}) = \mu(\mathcal{T}_x)$$

is known. Using the fact that $C(a,b) \ge a/(a+b+1)$ for $a \ge b$, we deduce that

$$1 \ge \lim_{\varepsilon\to 0} \prod_{i=0}^{h_n-1} C(a_i^{(n)}, b_i^{(n)}) \mathbf{1}_{b_i^{(n)} \le \varepsilon a_i^{(n)}} \ge \lim_{\varepsilon\to 0} \prod_{i=0}^{h_n-1} \frac{a_i^{(n)}}{a_i^{(n)}+b_i^{(n)}+1} \mathbf{1}_{b_i^{(n)} \le \varepsilon a_i^{(n)}} = 1,$$

from which (14) follows. □

3 Limits of trees and subtrees

Lemma 3.1. *Consider a sequence $((\mathcal{T}_n, \rho_n, \mu_n), n \in \mathbb{N})$ of deterministic rooted $\mathbb{R}$-trees which are subsets of a fixed compact metric space (E, d), such that $\mathcal{T}_n \to \mathcal{T}$, $\rho_n \to \rho$ and $\mu_n \to \mu$ for respectively the Hausdorff metric d_{H} on compact subsets of E, the metric d on E, and the Prokhorov metric d_{P} for Borel probability measures on E. Then, for any $z \in \mathcal{T}$, there exists a sequence $(z_n, n \in \mathbb{N})$ with $z_n \in T_n$ such that $z_n \to z$ and $\mu_n((\mathcal{T}_n)_{z_n}) \to \mu(\mathcal{T}_z)$.*

Proof. Write

$$\varepsilon_n := 2\max\big(d_{\mathrm{H}}(\mathcal{T}_n, \mathcal{T}), d(\rho_n, \rho), d_{\mathrm{P}}(\mu_n, \mu)\big),$$

this tends to 0 as n tends to infinity.

Let h be the height of z, for all n, x_n an element of $\mathcal{T}_n$ such that $d(z_n, x_n) \leq \varepsilon_n$, and y_n be the ancestor of x_n in $\mathcal{T}$ with height $\mathrm{ht}(x_n) - \varepsilon_n$, Notice that $d(y_n, z) \leq 3\varepsilon_n$

Let us show that $(\mathcal{T}_n)_{y_n}$ converges to T_z for the Hausdorff metric. We use the notation A^ε to denote the ε-enlargement of a subset A of E and will show that we have both $\mathcal{T}_z \subset (\mathcal{T}_n)^{\varepsilon_n}_{y_n}$ and $(\mathcal{T}_n)_{y_n} \subset \mathcal{T}^{\varepsilon'_n}_z$ with $\varepsilon'_n \to 0$.

Let $a \in \mathcal{T}_z$, we know that there is $b \in \mathcal{T}_n$ such that $d(a, b) \leq \varepsilon_n$, and we will show that b is in $(\mathcal{T}_n)_{y_n}$. By the triangle inequality we have $d(b, x_n) \leq d(b, a) + d(a, z) + d(z, x_n) \leq d(a, z) + 2\varepsilon_n$ and $d(a, z) \leq d(b, x_n) + 2\varepsilon_n$, hence $|d(b, x_n) - d(a, z)| \leq x\varepsilon'_n$. Similarly we obtain $|d(b, \rho_n) - d(a, \rho)| \leq 2\varepsilon_n$ and $|d(x_n, \rho_n) - d(z, \rho)| \leq 3\varepsilon_n$ Since $d(a, \rho) + d(a, z) + d(z, \rho)$, we deduce that $d(b, x_n) + d(x_n, \rho_n) - d(b, \rho_n) \leq 6\varepsilon_n$. Since $\mathcal{T}_n$ is an $\mathbb{R}$-tree, we can see that this difference is equal to twice the the distance from x_n to its most recent common ancestor with b is at most $3\varepsilon_n$, hence this most recent common ancestor is a descendant of y_n, hence $b \in (\mathcal{T}_n)_{y_n}$, hence $\mathcal{T}_z \subset (\mathcal{T}_n)^{\varepsilon_n}_{y_n}$.

Similarly, omitting the details, it can be seen that $(\mathcal{T}_n)_{y_n} \subset \mathcal{T}_{u_n}$, where u_n is an ancestor of z such that $d(u_n, z) \to 0$. However, since $\mathcal{T}$ is compact, we deduce that $\mathcal{T}_{u_n} \subset \mathcal{T}^{\varepsilon'_n}_z$, where ε'_n tends to 0, otherwise there would be a sequence with no subsequential limits.

We can now finish the proof. Letting $\eta_n = \max(\varepsilon_n, d_{\mathrm{H}}((\mathcal{T}_n)_{y_n}, T_z))$ for all n, we have $\mathcal{T}_z \subset (\mathcal{T}_n)^{\varepsilon_n}_{z'_n}$, we now finally define z_n as the ancestor of z'_n such that $\mathrm{ht}(z_n) = \mathrm{ht}(z'_n) - \eta_n$. The same argument as in the previous paragraph shows that $(\mathcal{T}_n)_{z_n}$ converges to $\mathcal{T}_z$ for the Hausdorff distance, hence $\eta'_n := \max(\eta_n, d_{\mathrm{H}}((\mathcal{T}_n)_{z_n}, T_z))$ tends to 0, and we have the following inclusions of sets:

$$\mathcal{T}_z \subset (\mathcal{T}_n)^{\eta_n}_{z'_n} \subset (\mathcal{T}_n)_{z_n} \subset \mathcal{T}^{\eta'_n}_z.$$

By the definition of the Prokhorov metric on measures, we then have

$$\mu(\mathcal{T}_z) \leq \mu_n((\mathcal{T}_n)_{z_n}) + \eta_n \leq \mu(\mathcal{T}^{\eta'_n}_z) + \eta_n + \eta'_n.$$

However, since $\mathcal{T}_z$ is a closed set, $\mu(\mathcal{T}^{\eta'_n}_z)$ converges to $\mu(\mathcal{T}_z)$, and by the sandwich theorem, so does $\mu_n((\mathcal{T}_n)_{z_n})$. □

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